



Birla Institute of Technology & Science, Pilani

Pilani Campus

AUGS/AGSR Division

GPIC 2026 | Gujarat Police Innovation Challenge

Module 01 — PRAHARI

Lectures Hackathon demo

Amit Dua, Arnav, Aria

September 2026



Outcomes

- State why PRAHARI does not replace departmental VMS
- Name the three layers that run in the PoC and the one that does not
- Recite the Sentinel ingest rules that the capture client obeys
- Walk the evaluation test: plate, track, watchlist alert, CSV



The deployed problem

- 26 departments run independent CCTV estates across about 1,000 km
- Mix of analog and IP, cloud and local NVR, 7-day vs 15-day retention
- VAHAN, SARTHI, eGujCop, AFIS, NAFIS already exist and the cameras do not match against them
- There is no statewide camera census



What we are not building

- A rip-and-replace of 26 VMS contracts
- Central recording of 80,000 cameras $\times$ 15 days on day one
- Face recognition as the evaluation demo
- A mock UI with no detection row in the database



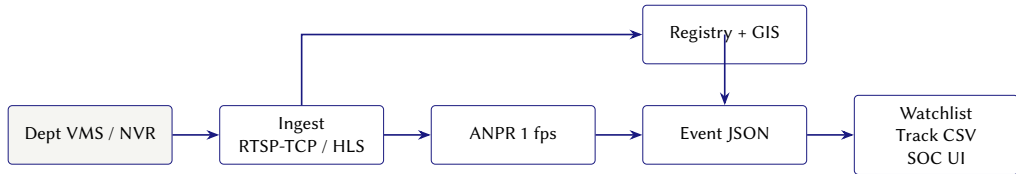
Chosen model

Layer	Official model	PoC
Census, GIS, health, gaps	1	now
Unified viewing + ANPR	2	now
Event bus, watchlist, track	thin 3	now
Central VMS recording	4	Phase-2, selected cameras

- A new vendor is one adapter on RTSP / HLS / WHEP / ONVIF
- Detection JSON does not change if the bus later becomes Kafka



Logical architecture



- Analytics samples 1 frame per second, not 25 fps GOP
- PoC bus is an in-process WebSocket; the JSON is the contract



Sentinel camera grid

- Catalogue: GET `http://<host>/api/ingest`. Camera ids change. The JSON is the contract
- RTSP `rtsp://<host>:8554/stream/<id>` for inference, TCP only
- HLS if port 8554 is blocked; WHEP for low-latency preview
- One second of video takes one second to arrive. There is no file download
- Host in use: `live.corp8.cloud` (ingest returned HTTP 502 on 01 Sep)



Integrator laws in the capture client

- Set `OPENCV_FFMPEG_CAPTURE_OPTIONS=rtsp_transport;tcp` before `import cv2`
- Event time is PTS (`CAP_PROP_POS_MSEC`), never arrival time, never declared FPS
- Reconnect backoff starts at 2 s and caps at 30 s
- Decoder warnings at join are logged, not fatal
- PTS rewind or gap > 5 s is a scene cut (feeds loop)
- Consume only. Do not publish. Do not `wget /stream/<id>`
- Tests: `tests/test_integrator_laws.py`



- Sampled frame → plate box → Tesseract → Indian normaliser
- Regex after stripping space and hyphen: `^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{3,4}$`
- Interface: `recognize(frame) -> {plate, plate_raw, confidence, crop}`
- Operator confirm writes confidence 1.0 so a readable plate cannot fail the demo
- Production swap (same function): YOLO plate detector + PaddleOCR



Watchlist and alerts

- In-memory plate set, $O(1)$ match, reload every 60 s
- Representative sources: VAHAN, SARTHI, eGujCop. No live ministry API in the PoC
- CRITICAL stolen/wanted (red); HIGH (amber); LOW (log)
- Same plate + same camera within 120 s increments a counter
- WebSocket /ws/alerts plus 5 s poll fallback
- WL-001 is GJ01AB1234 / STOLEN and cannot be deleted



Evaluation test: designated vehicle

Time	Camera	Location
06:12	CAM-VAL-001	Valsad NH-48 Toll
07:48	CAM-SUR-001	Surat Adajan Circle
09:05	CAM-AHD-004	Narol–Naroda Hwy
09:22	CAM-AHD-001	SG Highway Junction
09:51	CAM-GNR-003	Koba Circle
10:04	CAM-GNR-001	CH-0 Sector 21 GNR

- GET /api/track/GJ01AB1234 plus CSV download
- Live detections append; they do not replace the six seed points



Eighty thousand cameras (design target)

- About 45,000 public-domain cameras run analytics at 1 fps, 720p crop ≈ 80 KB
- Naive centre: $45,000 \times 80 \text{ KB/s} \approx 3.6 \text{ GB/s}$
- Five regions (Ahmedabad, Surat, Rajkot, Vadodara, Bhuj): $\approx 720 \text{ MB/s}$ each
- SOC wall pulls 16–64 HLS tiles. Everything else is metadata + crops
- These numbers are design targets, not measured laptop throughput



Cost of the intelligence plane

Item (statewide, per year)	INR
5 regional GPU nodes (3-yr amortised)	1.8–2.4 Cr
Object storage for crops + selected video	0.8–1.2 Cr
SWAN / leased bandwidth top-up	0.6–1.0 Cr
Integration + SOC ops (12 engineers)	1.5 Cr
Total intelligence plane	~5–6 Cr

- This is not the bill for replacing 26 VMS estates



Security in the PoC

- Roles: superadmin, soc_operator, dept_viewer, auditor
- home.viewer sees Home cameras only; private-permitted mall is hidden
- HLS/file tiles use a 60 s HMAC token. Raw RTSP never reaches the browser
- Private-permitted onboard requires consent=true
- Audit: onboard, watchlist edit, ack, report download, operator confirm



PoC mapped to evaluation

Evaluation item	Where it lives
Onboard gov + own feeds	/api/cameras, CSV, sync-catalogue
Unified viewing	map + tokenised /api/stream
ANPR + timestamps	/api/ingest/frame, confirm
Designated vehicle	/api/track/GJ01AB1234 + CSV
Watchlist + alert	/api/watchlist, /ws/alerts
80k readiness	HLD \$5 (design target) and \$10



Demo run (3 minutes)

- Open <http://127.0.0.1:8080>. Login judge
- Operations map: two pins, Dahod offline
- Vehicle Track: reconstruct GJ01AB1234, download CSV
- Alerts: CRITICAL queue, Ack writes audit
- Onboard: still ANPR or Operator confirm
- Gaps: short retention on Food and Civil Supplies
- GitHub: <https://github.com/amitduabits/PRAHARI>